CPE 381 Classwork 11: Fourier Series

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: October 07, 2026 (In-Class)

1. Consider the period impulse train shown in Figure 1

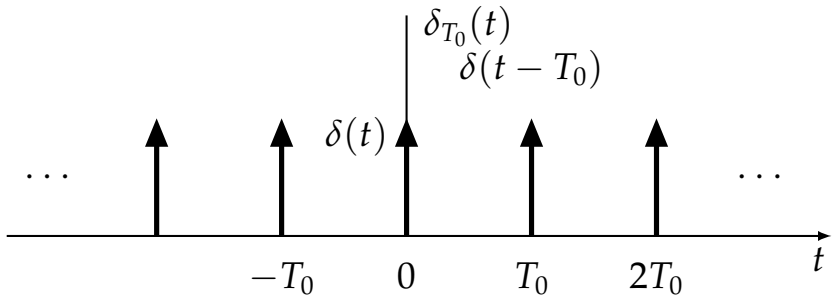


Figure 1: The periodic impulse train $\delta_{T_0}(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$.

Determine the complex exponential Fourier series of $\delta_{T_0}(t)$.

Answer

$$\delta_{T_0}(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$$

Note that the impulse function is an even function. Time period $T = T_0$

$$X_k = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t)\, e^{-jk\Omega_0 t}\, dt = \frac{1}{T_0} \int_{-\infty}^{\infty} \delta(t)\, e^{-jk\Omega_0 t}\, dt$$

We apply sifting property of δ function

$$X_k = \frac{1}{T_0} e^{-j \cdot k \cdot \Omega_0 \cdot 0} = \frac{1}{T_0} \times 1$$

Answer

So

$$\delta_{T_0}(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\Omega_0 t}$$

$$= \sum_{k=-\infty}^{\infty} \frac{1}{T_0} e^{jk \cdot \frac{2\pi}{T_0} t}$$

$$\Omega_0 = \frac{2\pi}{T_0}$$